EARTHSCAPE CLIMATE AGENCY

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Earthscape Climate Agency

Supervisor	Sir Wajahat	
Batch	2309f	
Project Title	Earthscape Climate Agency	
Serial no.	Enrollment no.	Student name
1.	1500503	M. Masoom
2.	1476532	Hamna Sarfaraz

Developer's Guide

Version: 1.0.0
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Platform: React (Vite) + FastAPI + MongoDB + Hadoop

1. GETTING STARTED

Requirements

- Node.js (v18 or above)
- Python 3.11+
- MongoDB
- Docker Desktop
- Hadoop
- Apache Hive
- Git
- Visual Studio Code

Installation

git clone https://github.com/repository/EarthScape.git

cd EarthScape

Frontend Setup

cd frontend

npm install

npm run dev

Backend Setup

cd backend

pip install -r requirements.txt

uvicorn main:app --reload

2. PROJECT STRUCTURE SNAPSHOT

```
frontend/
├── src/
│   ├── assets/
│   ├── components/
│   ├── features/
│   │   ├── alerts/
│   │   ├── auth/
│   │   ├── climate/
│   │   ├── dashboard/
│   │   ├── ingestion/
│   │   ├── jobs/
│   │   ├── support/
│   │   └── users/
│   ├── hooks/
│   ├── pages/
│   ├── services/
│   ├── store/
│   ├── styles/
│   └── utils/
│
backend/
├── app/
├── api/
├── models/
├── services/
├── database/
└── main.py
```

3. AUTHENTICATION

- Secure user login
- Session management
- Protected application routes
- Role-based user access
- FastAPI authentication middleware

Protected routes are handled through frontend route protection and backend API validation.

4. CORE MODULES

Dashboard

- Climate overview
- Statistics
- Interactive charts
- Recent activities

Climate Analytics

- Climate trend analysis
- Dataset visualization
- Report generation

Data Ingestion

- Upload climate datasets
- Dataset validation
- Storage in MongoDB

Alerts

- Climate notifications
- Threshold monitoring
- Alert history

Jobs

- Hadoop job monitoring
- Processing status
- Execution logs

Users

- User management
- Profile management

5. ADMIN CONTROLS

Administrator Features

- Manage users
- Monitor uploaded datasets
- View processing jobs
- Configure alerts
- Access system reports

Administrative routes are protected using authentication and authorization checks.

6. BIG DATA PROCESSING

EarthScape integrates Hadoop technologies for large-scale climate data processing.

Components

- Hadoop Distributed File System (HDFS)
- Apache Hive
- MapReduce
- MongoDB

Workflow

1. Upload dataset
2. Store data
3. Transfer to HDFS
4. Execute MapReduce jobs
5. Query results using Hive
6. Display analytics

7. API COMMUNICATION

Frontend communicates with the backend through REST APIs.

Responsibilities

- Authentication
- Climate data upload
- Dashboard statistics
- Alert retrieval
- User management

API requests are managed through reusable service modules.

8. SEARCH, FILTERS & DATA VISUALIZATION

Features include

- Climate data search
- Filtering datasets
- Interactive charts
- Dashboard summaries
- Analytical reports

Visualization updates dynamically after successful processing.

9. TESTING

Testing Types

- API Testing
- Component Testing
- Integration Testing
- User Interface Testing

Useful Commands

npm test

pytest

10. BUILD & DEPLOYMENT

Frontend

npm run build

Backend

uvicorn main:app

Docker

docker compose up --build

Deployment can be performed using Docker containers.

11. TROUBLESHOOTING

Common Commands

npm install

pip install -r requirements.txt

docker compose up

docker compose down

Verify

- MongoDB is running
- Docker services are active
- Hadoop services are configured
- Backend API is accessible
- Environment variables are correctly configured

12. FUTURE ENHANCEMENTS

- AI-based climate prediction
- Real-time weather APIs
- GIS map integration
- Machine Learning models
- Cloud deployment
- Mobile application
- Advanced reporting
- Multi-language support

END OF DEVELOPER'S GUIDE

THANK YOU!